STAT 312: Exam 1 Data Analysis

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1. Introduction

This report provides a descriptive analysis of Exam 1 scores from Sections 103, 104, and 105 of STAT 312.

There were **182 students** in the combined data set. Exam scores were recorded for **181 students**; one student had a missing score. The missing observation was excluded from the numerical and graphical analyses rather than being treated as a score of zero.

The objective is to use both **numerical summaries** and **graphical displays** to describe the center, variability, shape, and unusual observations in the Exam 1 data.

2. Overall Exam Performance

Table 1 summarizes Exam 1 performance across all three sections.

Statistic	**Value**
Students	182
Graded	181
Missing	1
Mean	33.96
Median	34.75
Standard deviation	4.18
Variance	17.51
Minimum	18.00
First quartile (Q_1)	32.00
Third quartile (Q_3)	37.00
Maximum	40.00
IQR	5.00

Table 1: Overall summary statistics for Exam 1.

The overall mean and median were

$$\bar{x} = 33.96 \qquad \text{and} \qquad \tilde{x} = 34.75.$$

The middle 50% of the scores were between

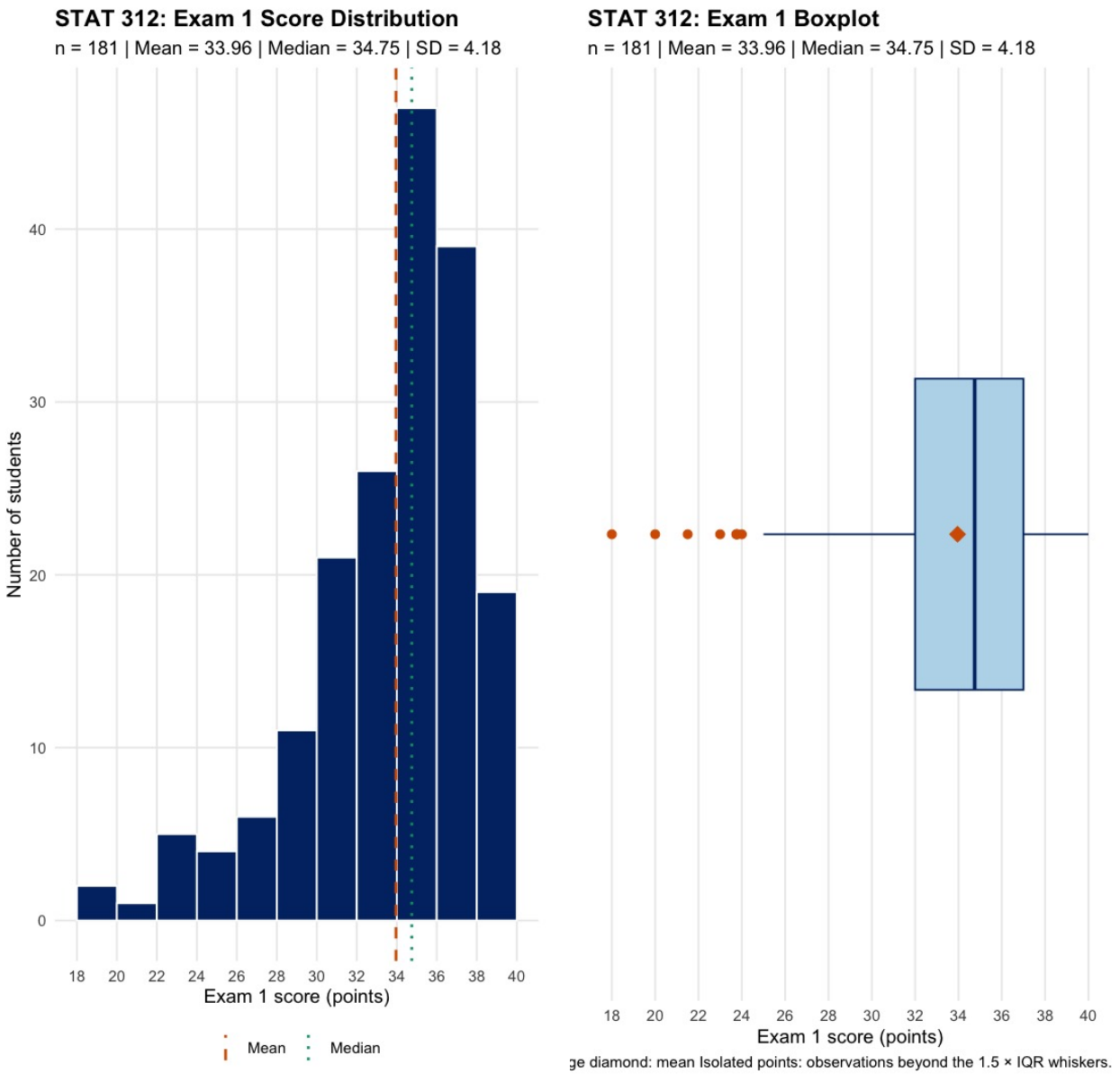
$$Q_1 = 32 \qquad \text{and} \qquad Q_3 = 37,$$

giving

$$IQR = Q_3 - Q_1 = 37 - 32 = 5.$$

The standard deviation was 4.18 points.

Graphical Analysis of All Sections



(a) Histogram of Exam 1 scores.

(b) Boxplot of Exam 1 scores.

Figure 1: Distribution of Exam 1 scores across all three sections.

Figure 1 presents two complementary views of the same data. The histogram shows that most students scored between approximately 30 and 40 points, with the greatest concentration of observations in the mid-to-upper 30s.

The distribution has a longer tail toward the lower scores. Consistent with this pattern,

$$\text{Mean} = 33.96 < \text{Median} = 34.75,$$

which suggests a **slightly left-skewed (negatively skewed)** distribution.

The boxplot reinforces this conclusion. The median lies above the mean, and several observations appear below the lower whisker. These points are identified as potential outliers using the usual $1.5 \times IQR$ criterion.

Thus, the histogram is particularly useful for examining the **shape** of the distribution, while the boxplot provides a compact summary of its **center, spread, and potential outliers**.

3. Comparison of Sections 103, 104, and 105

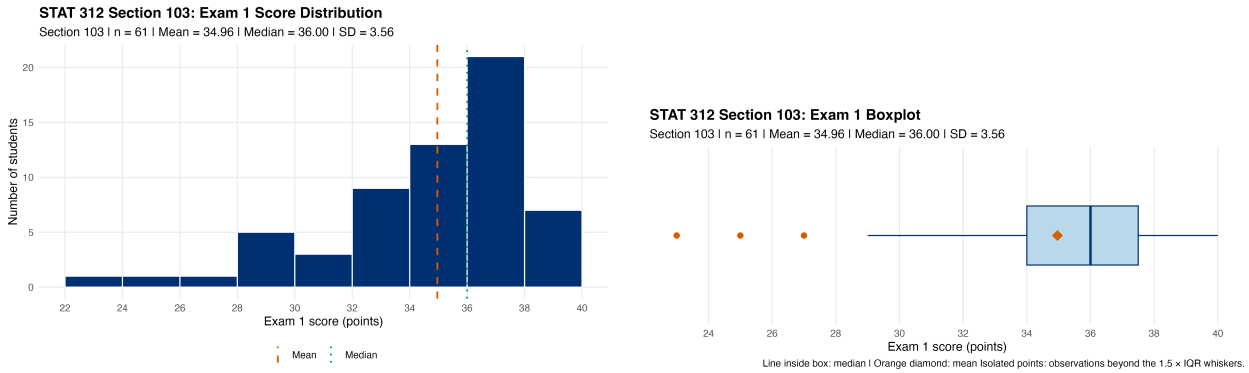
Table 2 compares the three sections.

Section	n	Mean	Median	SD	Min	Q_1	Q_3	Max	IQR
103	61	34.96	36.00	3.56	23.00	34.00	37.50	40.00	3.50
104	60	33.04	33.75	4.27	21.50	31.38	36.00	40.00	4.62
105	60	33.85	35.00	4.52	18.00	32.00	37.12	40.00	5.12

Table 2: Descriptive statistics for Exam 1 by section. Section 105 had one additional student with a missing Exam 1 score.

Although the three sections have broadly similar centers, there are noticeable differences in both center and variability.

Section 103



(a) Histogram. (b) Boxplot.

Figure 2: Exam 1 score distribution for Section 103.

Section 103 had the highest mean and median:

$$\bar{x}_{103} = 34.96, \qquad \tilde{x}_{103} = 36.00.$$

The histogram shows that most observations are concentrated toward the upper end of the score range. The lower tail consists of only a relatively small number of observations.

The boxplot identifies several potential low-end outliers. Nevertheless, the middle 50% of the scores are tightly concentrated between

$$Q_1 = 34.00 \qquad \text{and} \qquad Q_3 = 37.50.$$

Section 103 also has the smallest standard deviation,

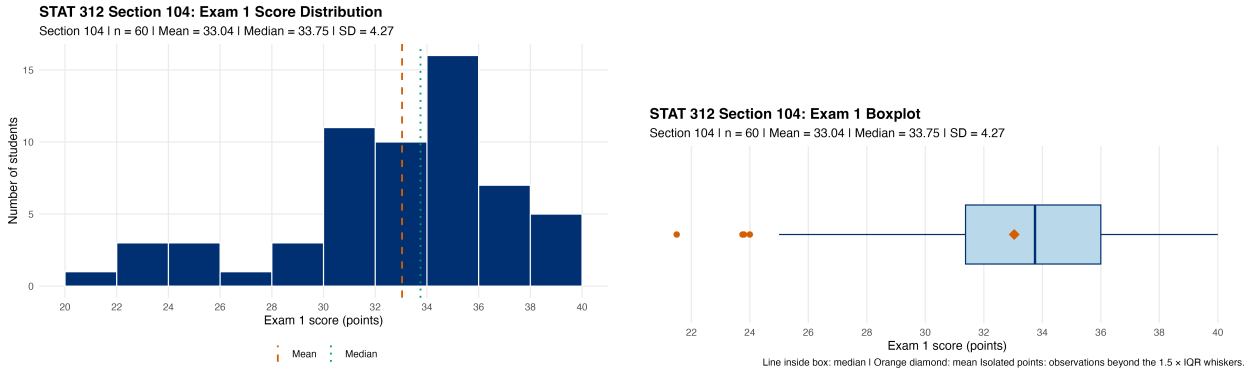
$$SD_{103} = 3.56,$$

and the smallest IQR,

$$IQR_{103} = 3.50.$$

Therefore, among the three sections, Section 103 has the **highest center and the least variability**.

Section 104



(a) Histogram. (b) Boxplot.

Figure 3: Exam 1 score distribution for Section 104.

For Section 104,

$$\bar{x}_{104} = 33.04, \qquad \tilde{x}_{104} = 33.75.$$

The histogram shows that most scores remain concentrated in the 30–40 point range, but several observations extend considerably farther toward the lower end.

The same pattern is visible in the boxplot, which identifies several potential low-score outliers.

Compared with Section 103, Section 104 has greater variability:

$$SD_{104} = 4.27 \qquad \text{and} \qquad IQR_{104} = 4.62.$$

Thus, Section 104 has a slightly lower center and a wider spread than Section 103.

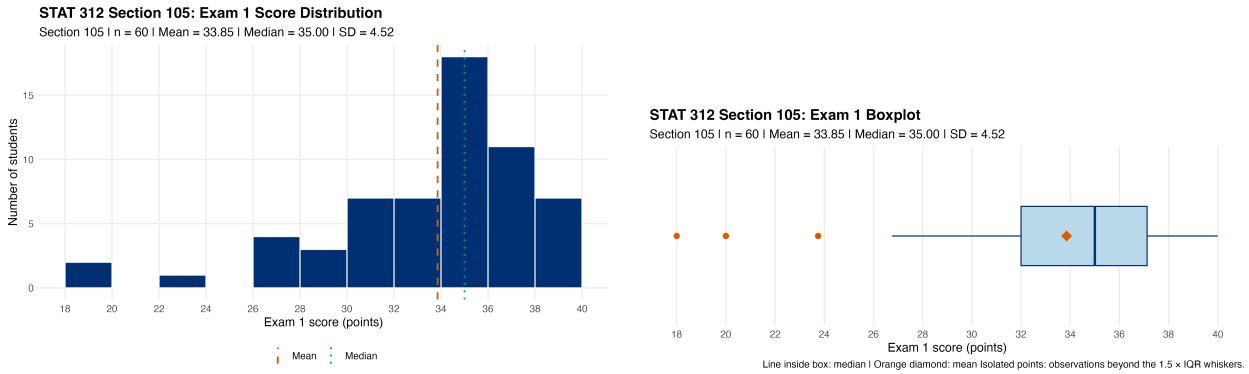
Section 105

For Section 105,

$$\bar{x}_{105} = 33.85, \qquad \tilde{x}_{105} = 35.00.$$

The histogram again shows substantial concentration in the 30–40 point range, together with a longer lower tail. The lowest recorded Exam 1 score among all three sections, 18, occurred in Section 105.

The boxplot identifies several relatively low observations as potential outliers.



(a) Histogram. (b) Boxplot.

Figure 4: Exam 1 score distribution for Section 105.

Section 105 has the largest standard deviation and IQR:

$$SD_{105} = 4.52, \qquad IQR_{105} = 5.12.$$

Thus, Section 105 demonstrates the **greatest variability** among the three sections.

4. Comparing the Three Distributions

The numerical and graphical results can now be considered together.

1. **Section 103 had the highest center.**

   Its mean and median were both larger than those of Sections 104 and 105.

2. **Section 103 also had the least variability.**

   It had both the smallest standard deviation and the smallest IQR.

3. **Section 105 had the greatest variability.**

   Although its mean was relatively close to the other sections, its standard deviation and IQR were the largest.

4. **All three distributions show some degree of lower-tail behavior.**

   In each section,

   $$\text{Mean} < \text{Median},$$

   and the boxplots identify unusually low observations.

5. **All three sections had a maximum score of 40.**

An important statistical lesson can be seen by comparing Sections 104 and 105. Their means are relatively close:

$$33.04 \qquad \text{versus} \qquad 33.85,$$

but their distributions are not identical. Section 105 has a larger standard deviation, larger IQR, and a substantially smaller minimum.

Therefore, comparing groups using only their means can hide important features of the data.

5. Lessons for Real Data Analysis

This analysis illustrates several fundamental principles of descriptive statistics.

- The **mean and median** describe the center of a distribution.
- The **standard deviation and IQR** describe variability.
- A **histogram** provides information about the shape, concentration, skewness, and possible gaps in the distribution.
- A **boxplot** provides a compact summary of the median, quartiles, spread, and potential outliers.
- Numerical summaries and graphical summaries should be interpreted **together**.
- When comparing multiple groups, we should examine both **center and variability**, rather than comparing means alone.

6. Final Summary

Across the three STAT 312 sections, the overall Exam 1 mean was

$$\boxed{\bar{x} = 33.96},$$

with median

$$\boxed{\tilde{x} = 34.75}.$$

Most scores were concentrated toward the upper part of the observed range, while a relatively small number of low scores produced a longer lower tail.

Section 103 had the highest center and smallest variability, whereas Section 105 exhibited the greatest variability.

The main lesson from this real-data example is:

A good statistical analysis should examine the data both numerically and graphically.

No single statistic or graph tells the entire story.

A R Code for Exam 1 Data Analysis

The following R code was used to produce the descriptive statistics, histograms, and boxplots presented in this report. File locations have been omitted for privacy and portability. Interested users should replace the placeholder directory with the location of their own data files.

A.1 Analysis of All Sections Combined

```

##########################################
###### STAT 312: EXAM 01 ANALYSIS ######
############## ALL SECTIONS ##############
##########################################

# Run once if needed:
# install.packages(c("dplyr", "ggplot2"))

library(dplyr)
library(ggplot2)

# ------------------------------------------------------------
# Set the folder containing the data
# ------------------------------------------------------------

# Replace this with the location of your own data files.
folder <- "PATH/TO/GRADES"

# ------------------------------------------------------------
# Read data
# ------------------------------------------------------------

grades <- read.csv(
  file.path(folder, "Exam01.csv"),
  check.names = FALSE,
  stringsAsFactors = FALSE,
  na.strings = c("", "NA"),
  fileEncoding = "UTF-8-BOM"
)

# ------------------------------------------------------------
# Clean data
# ------------------------------------------------------------

# Remove metadata and the test account before converting scores.

exam <- grades |>
  mutate(Student = trimws(Student)) |>
  filter(
    !is.na(Student),
    Student != "",
    Student != "Points Possible",
    !grepl(
      "^Student,[[:space:]]*Test$",
      Student,
      ignore.case = TRUE
    )
  ) |>
  mutate(
    Score = as.numeric(`EXAM 01 (780600)`)
  )

# ------------------------------------------------------------
# Summary statistics
# ------------------------------------------------------------

```

```r
exam_summary <- exam |>
  summarise(
    Students = n(),
    Graded = sum(!is.na(Score)),
    Missing = sum(is.na(Score)),
    Mean = mean(Score, na.rm = TRUE),
    Median = median(Score, na.rm = TRUE),
    SD = sd(Score, na.rm = TRUE),
    Variance = var(Score, na.rm = TRUE),
    Minimum = min(Score, na.rm = TRUE),
    Q1 = unname(
      quantile(Score, 0.25, na.rm = TRUE)
    ),
    Q3 = unname(
      quantile(Score, 0.75, na.rm = TRUE)
    ),
    Maximum = max(Score, na.rm = TRUE),
    IQR = IQR(Score, na.rm = TRUE)
  )

print(
  as.data.frame(
    round(exam_summary, 2)
  )
)

# ------------------------------------------------------------
# Use only recorded scores for plots
# ------------------------------------------------------------

exam_scored <- exam |>
  filter(!is.na(Score))

mean_score <- mean(exam_scored$Score)

median_score <- median(exam_scored$Score)

plot_subtitle <- sprintf(
  "n = %d | Mean = %.2f | Median = %.2f | SD = %.2f",
  nrow(exam_scored),
  mean_score,
  median_score,
  sd(exam_scored$Score)
)

# ------------------------------------------------------------
# Histogram
# ------------------------------------------------------------

histogram <- ggplot(
  exam_scored,
  aes(x = Score)
) +

  geom_histogram(
    binwidth = 2,
    boundary = 0,
    fill = "#003071",
    color = "white"
  ) +

  geom_vline(
    aes(
      xintercept = mean_score,
      linetype = "Mean"
    ),
    color = "#D55E00",
    linewidth = 0.9
```

```
  ) +

  geom_vline(
    aes(
      xintercept = median_score,
      linetype = "Median"
    ),
    color = "#009E73",
    linewidth = 0.9
  ) +

  scale_linetype_manual(
    name = NULL,
    values = c(
      "Mean" = "dashed",
      "Median" = "dotted"
    )
  ) +

  scale_x_continuous(
    breaks = scales::breaks_width(2)
  ) +

  labs(
    title = "STAT 312: Exam 1 Score Distribution",
    subtitle = plot_subtitle,
    x = "Exam 1 score (points)",
    y = "Number of students"
  ) +

  theme_minimal(
    base_size = 13
  ) +

  theme(
    plot.title = element_text(
      face = "bold"
    ),
    legend.position = "bottom",
    panel.grid.minor = element_blank()
  )

print(histogram)

# ------------------------------------------------------------
# Horizontal boxplot
# ------------------------------------------------------------

boxplot <- ggplot(
  exam_scored,
  aes(
    x = Score,
    y = ""
  )
) +

  geom_boxplot(
    width = 0.4,
    fill = "#B9D9EB",
    color = "#003071",
    outlier.color = "#D55E00",
    outlier.size = 2.5
  ) +

  stat_summary(
    fun = mean,
    geom = "point",
    shape = 23,
```

```
    size = 3.5,
    fill = "#D55E00",
    color = "#D55E00"
  ) +

  scale_x_continuous(
    breaks = scales::breaks_width(2)
  ) +

  labs(
    title = "STAT 312: Exam 1 Boxplot",
    subtitle = plot_subtitle,
    x = "Exam 1 score (points)",
    y = NULL,
    caption = paste(
      "Line inside box: median | Orange diamond: mean",
      "Isolated points: observations beyond the 1.5 x IQR whiskers."
    )
  ) +

  theme_minimal(
    base_size = 13
  ) +

  theme(
    plot.title = element_text(
      face = "bold"
    ),
    panel.grid.major.y = element_blank(),
    panel.grid.minor = element_blank()
  )

print(boxplot)

# ------------------------------------------------------------
# Save results
# ------------------------------------------------------------

write.csv(
  exam_summary,
  file.path(
    folder,
    "Exam01_Summary.csv"
  ),
  row.names = FALSE
)

ggsave(
  file.path(
    folder,
    "Exam01_Histogram.png"
  ),
  histogram,
  width = 9,
  height = 5.5,
  dpi = 300
)

ggsave(
  file.path(
    folder,
    "Exam01_Boxplot.png"
  ),
  boxplot,
  width = 9,
  height = 4,
  dpi = 300
)
```

Listing 1: R code for the combined Exam 1 analysis.

A.2 Analysis by Section

The following function applies the same analysis separately to Sections 103, 104, and 105. Using a function and loop avoids repeating the same analysis code for each section.

```
1
2  ########################################
3  ###### STAT 312: EXAM 01 ANALYSIS ######
4  ########## BY SECTION: 103-105 ##########
5  ########################################
6
7  library(dplyr)
8  library(ggplot2)
9
10 # ------------------------------------------------------------
11 # Folder setup
12 # ------------------------------------------------------------
13
14 # Replace this with the location of your own data files.
15 folder <- "PATH/TO/GRADES"
16
17 # Create a folder for section-level results.
18
19 output_folder <- file.path(
20   folder,
21   "Section Analysis"
22 )
23
24 if (!dir.exists(output_folder)) {
25   dir.create(output_folder)
26 }
27
28 # ============================================================
29 # Function to analyze one section
30 # ============================================================
31
32 analyze_exam_section <- function(
33     section,
34     folder,
35     output_folder,
36     exam_column = "EXAM 01 (780600)"
37 ) {
38
39   # ----------------------------------------------------------
40   # File name
41   # ----------------------------------------------------------
42
43   file_name <- paste0(
44     section,
45     "-Exam01.csv"
46   )
47
48   file_path <- file.path(
49     folder,
50     file_name
51   )
52
53   # ----------------------------------------------------------
54   # Read Canvas CSV
55   # ----------------------------------------------------------
56
57   grades <- read.csv(
```

```
    file_path,
    check.names = FALSE,
    stringsAsFactors = FALSE,
    na.strings = c("", "NA"),
    fileEncoding = "UTF-8-BOM"
  )

  # ------------------------------------------------------------
  # Clean data
  # ------------------------------------------------------------

  exam <- grades |>
    mutate(
      Student = trimws(Student)
    ) |>
    filter(
      !is.na(Student),
      Student != "",
      Student != "Points Possible",
      !grepl(
        "^Student,[[:space:]]*Test$",
        Student,
        ignore.case = TRUE
      )
    ) |>
    mutate(
      Score = suppressWarnings(
        as.numeric(
          .data[[exam_column]]
        )
      )
    )

  # ------------------------------------------------------------
  # Summary statistics
  # ------------------------------------------------------------

  exam_summary <- exam |>
    summarise(
      Section = section,
      Students = n(),
      Graded = sum(!is.na(Score)),
      Missing = sum(is.na(Score)),
      Mean = mean(
        Score,
        na.rm = TRUE
      ),
      Median = median(
        Score,
        na.rm = TRUE
      ),
      SD = sd(
        Score,
        na.rm = TRUE
      ),
      Variance = var(
        Score,
        na.rm = TRUE
      ),
      Minimum = min(
        Score,
        na.rm = TRUE
      ),
      Q1 = unname(
        quantile(
          Score,
          0.25,
          na.rm = TRUE
```

```
      )
    ),
    Q3 = unname(
      quantile(
        Score,
        0.75,
        na.rm = TRUE
      )
    ),
    Maximum = max(
      Score,
      na.rm = TRUE
    ),
    IQR = IQR(
      Score,
      na.rm = TRUE
    )
  )

# ------------------------------------------------------------
# Print summary
# ------------------------------------------------------------

cat(
  "\n",
  "========================================\n",
  "STAT 312 - SECTION ",
  section,
  "\n",
  "========================================\n",
  sep = ""
)

print(
  exam_summary |>
    mutate(
      across(
        where(is.numeric),
        ~ round(.x, 2)
      )
    )
)

# ------------------------------------------------------------
# Recorded scores only
# ------------------------------------------------------------

exam_scored <- exam |>
  filter(
    !is.na(Score)
  )

mean_score <- mean(
  exam_scored$Score
)

median_score <- median(
  exam_scored$Score
)

sd_score <- sd(
  exam_scored$Score
)

plot_subtitle <- sprintf(
  paste0(
    "Section %s | n = %d | ",
    "Mean = %.2f | Median = %.2f | SD = %.2f"
```

```
    ),
    section,
    nrow(exam_scored),
    mean_score,
    median_score,
    sd_score
  )

  # ============================================================
  # Histogram
  # ============================================================

  histogram <- ggplot(
    exam_scored,
    aes(x = Score)
  ) +

    geom_histogram(
      binwidth = 2,
      boundary = 0,
      fill = "#003071",
      color = "white"
    ) +

    geom_vline(
      aes(
        xintercept = mean_score,
        linetype = "Mean"
      ),
      color = "#D55E00",
      linewidth = 0.9
    ) +

    geom_vline(
      aes(
        xintercept = median_score,
        linetype = "Median"
      ),
      color = "#009E73",
      linewidth = 0.9
    ) +

    scale_linetype_manual(
      name = NULL,
      values = c(
        "Mean" = "dashed",
        "Median" = "dotted"
      )
    ) +

    scale_x_continuous(
      breaks = scales::breaks_width(2)
    ) +

    labs(
      title = paste0(
        "STAT 312 Section ",
        section,
        ": Exam 1 Score Distribution"
      ),
      subtitle = plot_subtitle,
      x = "Exam 1 score (points)",
      y = "Number of students"
    ) +

    theme_minimal(
      base_size = 13
    ) +
```

```
    theme(
      plot.title = element_text(
        face = "bold"
      ),
      legend.position = "bottom",
      panel.grid.minor = element_blank()
    )

print(histogram)

# ============================================================
# Horizontal boxplot
# ============================================================

boxplot <- ggplot(
  exam_scored,
  aes(
    x = Score,
    y = ""
  )
) +

  geom_boxplot(
    width = 0.4,
    fill = "#B9D9EB",
    color = "#003071",
    outlier.color = "#D55E00",
    outlier.size = 2.5
  ) +

  stat_summary(
    fun = mean,
    geom = "point",
    shape = 23,
    size = 3.5,
    fill = "#D55E00",
    color = "#D55E00"
  ) +

  scale_x_continuous(
    breaks = scales::breaks_width(2)
  ) +

  labs(
    title = paste0(
      "STAT 312 Section ",
      section,
      ": Exam 1 Boxplot"
    ),
    subtitle = plot_subtitle,
    x = "Exam 1 score (points)",
    y = NULL,
    caption = paste(
      "Line inside box: median | Orange diamond: mean",
      "Isolated points: observations beyond the 1.5 x IQR whiskers."
    )
  ) +

  theme_minimal(
    base_size = 13
  ) +

  theme(
    plot.title = element_text(
      face = "bold"
    ),
    panel.grid.major.y = element_blank(),
```

```
      panel.grid.minor = element_blank()
    )

  print(boxplot)

  # ============================================================
  # Save results for this section
  # ============================================================

  write.csv(
    exam_summary,
    file.path(
      output_folder,
      paste0(
        "Exam01_Section",
        section,
        "_Summary.csv"
      )
    ),
    row.names = FALSE
  )

  ggsave(
    file.path(
      output_folder,
      paste0(
        "Exam01_Section",
        section,
        "_Histogram.png"
      )
    ),
    histogram,
    width = 9,
    height = 5.5,
    dpi = 300
  )

  ggsave(
    file.path(
      output_folder,
      paste0(
        "Exam01_Section",
        section,
        "_Boxplot.png"
      )
    ),
    boxplot,
    width = 9,
    height = 4,
    dpi = 300
  )

  # ------------------------------------------------------------
  # Return results
  # ------------------------------------------------------------

  return(
    list(
      summary = exam_summary,
      data = exam,
      scored_data = exam_scored,
      histogram = histogram,
      boxplot = boxplot
    )
  )
}

# ============================================================
```

```r
# Run function for all three sections
# ============================================================

sections <- c(
  "103",
  "104",
  "105"
)

results <- lapply(
  sections,
  function(sec) {

    analyze_exam_section(
      section = sec,
      folder = folder,
      output_folder = output_folder
    )

  }
)

names(results) <- sections

# ============================================================
# Combine summary statistics
# ============================================================

all_section_summary <- bind_rows(
  lapply(
    results,
    function(x) x$summary
  )
)

cat(
  "\n\n",
  "========================================\n",
  "SUMMARY: ALL THREE SECTIONS\n",
  "========================================\n",
  sep = ""
)

print(
  all_section_summary |>
    mutate(
      across(
        where(is.numeric),
        ~ round(.x, 2)
      )
    )
)

# ============================================================
# Save combined summary
# ============================================================

write.csv(
  all_section_summary,
  file.path(
    output_folder,
    "Exam01_All_Sections_Summary.csv"
  ),
  row.names = FALSE
)
```

Listing 2: R function and loop for section-level Exam 1 analyses.

A.3 Reproducibility Note

The analysis was conducted in R using the `dplyr` and `ggplot2` packages. The code is included to illustrate how the numerical summaries and graphical displays presented in this report can be reproduced from the original data.

Student-identifying information is not displayed in this report, and the local file paths used during the analysis have been removed from the reproduced code.